

Experiment No. 4

Aim:

To design a gas detection system using Arduino Uno, MQ-5 gas sensor, LED, resistor, and connecting wires to detect harmful gases like LPG and methane and provide a visual indication using an LED.

Explanation:

Components and Their Detailed Description

1. Arduino Uno

- **Description:** A microcontroller board based on the ATmega328P, commonly used for embedded systems and DIY electronics projects.
- **Features:**
 - 14 digital input/output pins (6 PWM outputs).
 - 6 analog inputs.
 - Operates at 5V with a clock speed of 16 MHz.
- **Function:**
 - Acts as the central control unit, processing sensor data and controlling output devices.
 - Reads input from the MQ-5 sensor and determines if the gas level is dangerous.
 - Controls the LED to turn ON/OFF based on the gas concentration.

2. MQ-5 Gas Sensor

- **Description:** A semiconductor gas sensor used to detect combustible gases like LPG, methane, and natural gas.
- **Features:**
 - Operates on 5V power supply.
 - Provides both digital and analog output.
 - High sensitivity to LPG, propane, and methane.

- **Function:**
 - Detects gas concentration in the surrounding air.
 - Outputs an analog voltage corresponding to the gas level.
 - Sends the signal to the Arduino for further processing.

3. LED Light

- **Description:** A Light Emitting Diode (LED) that glows when an electric current passes through it.
- **Function:**
 - Serves as a visual alert system when a high gas concentration is detected.
 - Controlled by the Arduino using a digital output pin.
 - Helps indicate system status (ON/OFF conditions).

4. Resistor

- **Description:** A passive electrical component that limits current flow.
- **Function:**
 - Used to protect the LED from excess current.
 - Helps stabilize the circuit by providing controlled voltage and current distribution.

5. Connecting Wires

- **Description:** Electrical conductors used to connect different components.
- **Function:**
 - Provide connections between the Arduino, MQ-5 sensor, LED, and power supply.
 - Enable smooth transmission of electrical signals.

Working of the Arduino-Based Gas Detection System

1. Powering the System

- The Arduino Uno and MQ-5 sensor are powered using a **5V power supply**.

- The sensor starts detecting the gas levels in the surrounding air.

2. Reading Sensor Data

- The **MQ-5 sensor outputs an analog voltage** that varies with gas concentration.
- The Arduino reads this signal from **analog input pin A0**.

3. Processing the Data in Arduino

- The Arduino compares the sensor reading with a **predefined threshold**.
- If the gas level is **below the threshold**, the LED remains **OFF**.
- If the gas level **exceeds the threshold**, the LED turns **ON** as a warning signal.

4. Output Indication

- **No gas detected → LED remains OFF.**
- **High gas concentration → LED turns ON** to alert the user.
- Additional components like a buzzer or display can be added for enhanced alert mechanisms.

Conclusion

This project successfully demonstrates a simple gas leakage detection system using Arduino. The MQ-5 sensor detects gas levels, and an LED provides a visual alert when dangerous concentrations are detected. This system can be further enhanced by integrating a buzzer or a GSM module to send alerts.